SHULL SCIENCE · JAMES A. GARFIELD LOCAL SCHOOLS

UNIT 2 TEST C: WRITTEN

UNIT 2 TEST · DAY 2 · WRITTEN · VERSION C · TEACHER KEY

NAME DATE PERIOD SCORE / 14

You may use a periodic table. Write every configuration in order of filling. In the orbital diagram, each electron gets one arrow.

Part 1 Longhand configurations

8 points

Number 1 also needs its orbital diagram and the number of half-filled orbitals.

1 — 4 pts

16 **S** Sulfur 32.06

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$ **Half-filled orbitals** 2

Orbital diagram



2 — 2 pts

28 **Ni** Nickel 58.69

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$

3 — 2 pts

35 **Br** Bromine 79.90

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$

1 is 4 pts, configuration 2, diagram 1, half-filled count 1. 2 and 3 are 2 pts each, 1 off for one wrong sublevel.

Part 2 Noble-gas shorthand

6 points

4

24 **Cr** Chromium 52.00

$[Ar]\ 4s^1\ 3d^5$

5

76 **Os** Osmium 190.23

$[Xe]\ 6s^2\ 4f^{14}\ 5d^6$

6

49 **In** Indium 114.82

$[Kr]\ 5s^2\ 4d^{10}\ 5p^1$

2 pts each, noble-gas core 1 and the rest in order 1. Chromium is the exception, so the predicted $[Ar]\ 4s^2\ 3d^4$ earns 1.